

034IN - FONDAMENTI DI AUTOMATICA
- FUNDAMENTALS OF AUTOMATIC CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION
A.Y. 2024/2025
SOLUTION

June 20 2025

First and Last Name:

Group: Group A

Exercise: Two Exercises

Available Time Frame: 60 minutes

Important Notes:

- Write your answers on a single white sheet of paper using a black pen.
- Do not write in blue ink or pencil.
- Do not hand in additional sheets.
- The clarity and precision of your answers will be subject to evaluation.

I declare that the answers to this exercise are my own and only my own work and that I have not consulted with others.

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CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION – A.Y. 2024/2025
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First and Last Name:

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Exercise: Exercise 1 - 2 Questions

Available Time Frame: 60 minutes

Exercise 1

Consider the block diagram in the figure:

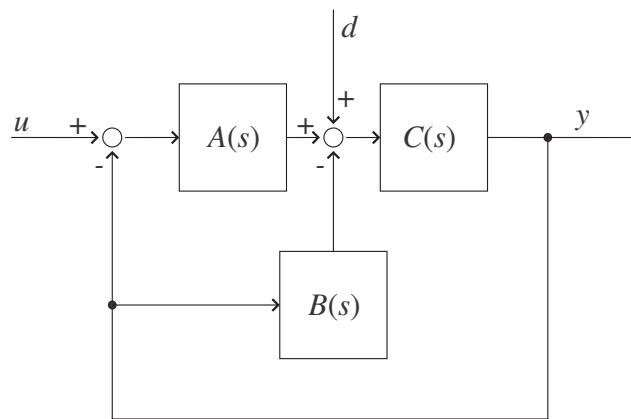


Figure 1: Block scheme

where the transfer functions $A(s)$, $B(s)$, and $C(s)$ assume the following expressions

$$A(s) = \frac{3}{s-10}, \quad B(s) = A(s), \quad C(s) = \frac{2s}{3s+1} \quad (1)$$

Question 1.1. Determine a **realisation using state equations** for the overall system.

Question 1.2. Analyse the stability of the overall system, just exploiting the structure and the topology of the block scheme depicted in Fig. 1.

First and Last Name:

Group: Group A

Exercise: Exercise 2 - 1 Question

Available Time Frame: 60 minutes

Exercise 2

Consider the continuous-time controller described by means of the transfer function:

$$C(s) = 5.0 \cdot \frac{1 + s}{(81 + 18s + s^2)}$$

By controlling an appropriate process $G(s)$ [whose expression is not necessary to know for solving the exercise], this controller allows the following performance to be obtained:

- phase margin $\varphi_m = 30^\circ$
- crossover angular frequency $\omega_c = 4.5 \text{ rad/sec}$

Question 2.1. Discretise the continuous-time controller $C(s)$, using the approximate ‘Tustin’ technique, choosing the sampling period appropriately, to ensure compliance with the fundamental sampling theorem and to estimate the maximum value for the decrease in the phase margin equal to:

$$|\delta_\varphi| \leq 1^\circ$$

Motivate the choices you make.

EXERCISE 1 – A SOLUTION PROPOSAL

Answer 1.1

Given the block scheme, put in evidence local input, output, and internal state variables associated with each dynamic elementary block in the scheme, as in the following Figure:

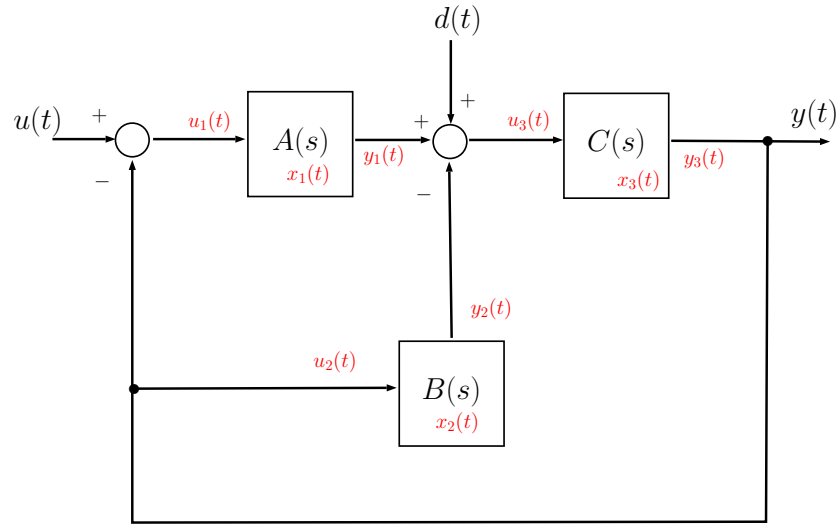
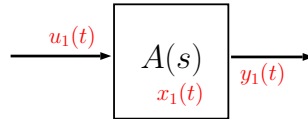


Figure 2: The block scheme, with in evidence local inputs, outputs and internal state variables (in red).

After that, consider each elementary dynamic block as a standalone, and determine a realisation in state space form for each of them, separately.

Realisation in state equation for the block with transfer function $A(s)$

- Consider only the following portion of the block scheme



where

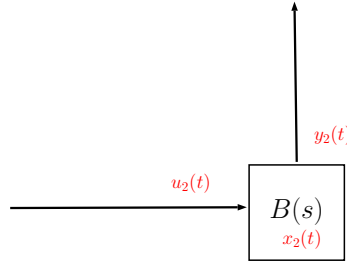
$$A(s) = \frac{3}{s - 10}$$

- Write the state equations using the “Observation Form” (refer to Part 4, page 37 of the course material):

$$\begin{cases} \dot{x}_1(t) &= +10x_1(t) + 3u_1(t) \\ y_1(t) &= x_1(t) \end{cases}$$

Realisation in state equation for the block with transfer function $B(s)$

- Consider only the following portion of the block scheme



where

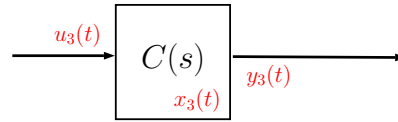
$$B(s) = A(s) = \frac{3}{s-10}$$

- Write the state equations using the “Observation Form” (refer to Part 4, page 37 of the course material):

$$\begin{cases} \dot{x}_2(t) &= +10x_2(t) + 3u_2(t) \\ y_2(t) &= x_2(t) \end{cases}$$

Realisation in state equation for the block with transfer function $C(s)$

- Consider only the following portion of the block scheme



where

$$C(s) = \frac{2s}{3s+1}$$

- Write the state equations using the “Observation Form”. Pay attention: the transfer function $C(s)$ is not strictly proper (refer to Part 4, page 38 of the course material):

$$C(s) = \frac{2}{3} - \frac{\frac{2}{3}}{3s+1} = \frac{2}{3} - \frac{\frac{2}{9}}{s+\frac{1}{3}} \implies \begin{cases} \dot{x}_3(t) &= -\frac{1}{3}x_3(t) - \frac{2}{9}u_3(t) \\ y_3(t) &= x_3(t) + \frac{2}{3}u_3(t) \end{cases}$$

Auxiliary equations describing the block scheme interconnections

Consider the block scheme depicted in Fig. 2: by analysing the scheme, you may write some mathematical expressions, describing the interconnections between the blocks inside the diagram:

$$\begin{cases} u_1(t) &= u(t) - y(t) \\ u_2(t) &= y(t) \\ u_3(t) &= d(t) + y_1(t) - y_2(t) \\ y_3(t) &= y(t) \end{cases}$$

Putting the equations altogether You should manipulate the resulting set of equations to obtain a subset of equations, containing only the three state variables, the inputs $u(t)$, $d(t)$, and the output $y(t)$.

$$\left\{ \begin{array}{lcl} \dot{x}_1(t) & = & +10x_1(t) + 3u_1(t) \\ \dot{x}_2(t) & = & +10x_2(t) + 3u_2(t) \\ \dot{x}_3(t) & = & -\frac{1}{3}x_3(t) - \frac{2}{9}u_3(t) \\ y_1(t) & = & x_1(t) \\ y_2(t) & = & x_2(t) \\ y_3(t) & = & x_3(t) + \frac{2}{3}u_3(t) \\ u_1(t) & = & u(t) - y(t) \\ u_2(t) & = & y(t) \\ u_3(t) & = & d(t) + y_1(t) - y_2(t) \\ y(t) & = & y_3(t) \end{array} \right. \Rightarrow \left\{ \begin{array}{lcl} \dot{x}_1(t) & = & +8x_1(t) + 2x_2(t) - 3x_3(t) + 3u(t) - 2d(t) \\ \dot{x}_2(t) & = & +2x_1(t) + 8x_2(t) + 3x_3(t) + 2d(t) \\ \dot{x}_3(t) & = & -\frac{2}{9}x_1(t) + \frac{2}{9}x_2(t) - \frac{1}{3}x_3(t) - \frac{2}{9}d(t) \\ y(t) & = & \frac{2}{3}x_1(t) - \frac{2}{3}x_2(t) + x_3(t) + \frac{2}{3}d(t) \end{array} \right.$$

In conclusion, the system matrices are

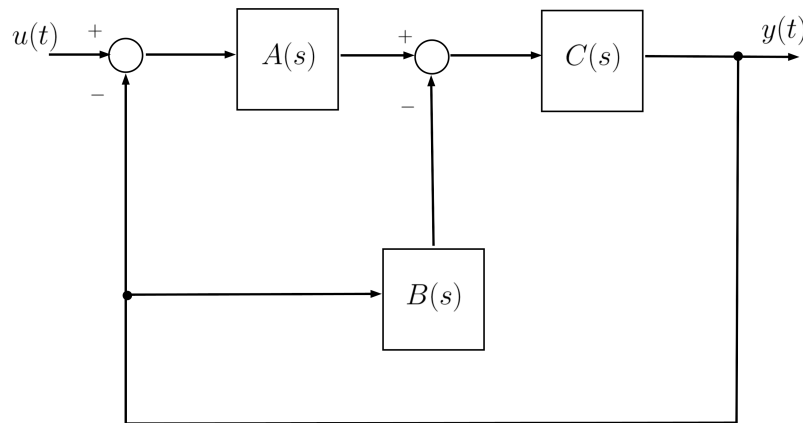
$$\begin{aligned} A &= \begin{bmatrix} +8 & +2 & -3 \\ +2 & +8 & +3 \\ -\frac{2}{9} & +\frac{2}{9} & -\frac{1}{3} \end{bmatrix}, & B &= \begin{bmatrix} +3 & -2 \\ 0 & +2 \\ 0 & -\frac{2}{9} \end{bmatrix} \\ C &= \begin{bmatrix} \frac{2}{3} & -\frac{2}{3} & +1 \end{bmatrix}, & D &= \begin{bmatrix} 0 & \frac{2}{3} \end{bmatrix} \end{aligned}$$

Answer 1.2

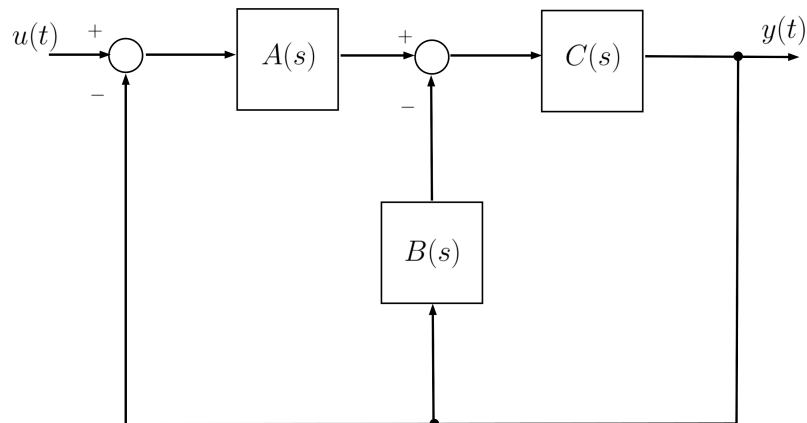
You are asked to analyse the stability of the overall system, exploiting the structure and the topology of the block scheme in Fig. 1. Recall **Part 6** of the course material: if you find common factors in the context of block-scheme reduction, and those factors involve unstable terms, then the overall system is unstable (recall **Part 6**, pages 3-6, and pages 9-12).

The system in Fig. 1 has two inputs: $u(t)$ and $d(t)$. You have to analyse the structure of the block scheme, looking for common unstable terms involving blocks in series or parallel interconnections. It is easy to point out such common terms, considering only one input at a time:

- Let's analyse the block scheme, with only the input $u(t)$ active (i.e., the second input $d(t)$ is forced to be null).

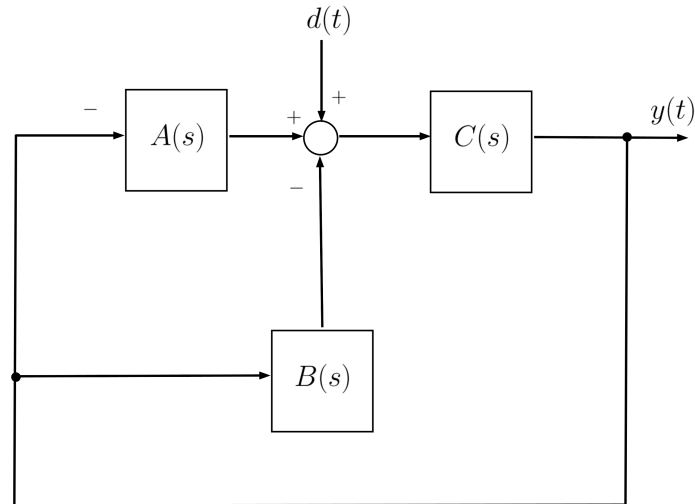


It's convenient to redraw the scheme as follows:

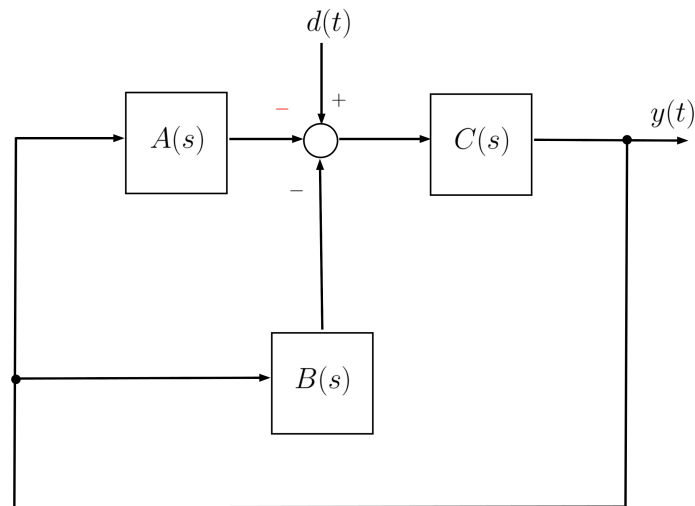


- Note: there are only feedback interconnections. There are no common terms in series or parallel interconnections (there are no series or parallel interconnections at all). You cannot determine the overall system's stability by analysing the block scheme topology in this case.

- Let's analyse the block scheme, with only the input $d(t)$ active (i.e., the first input $u(t)$ is forced to be null).



It's convenient to redraw the scheme as follows:



- Now it is clear that, this time, the blocks with transfer functions $A(s)$ and $B(s)$ are connected in parallel. (recall **Part 6, page 11** of the course material). The common term in this parallel interconnection is related to an unstable pole: $s - 10$ at the denominator in both transfer functions.
- **Concluding**, there is an hidden internal unstable dynamics, associated with the term $s - 10$: the **overall system** is **unstable**, and the unstable pole is $p = +10$.

As a final check, consider the A matrix in the state-equations representation of the overall system

$$A = \begin{bmatrix} +8 & +2 & -3 \\ +2 & +8 & +3 \\ -\frac{2}{9} & +\frac{2}{9} & -\frac{1}{3} \end{bmatrix}$$

and compute the eigenvalues of A :

$$\det(\lambda I - A) = s^3 - \frac{47}{3}s^2 + \frac{160}{3}s + \frac{100}{3} = 0 \quad \Longleftrightarrow \quad (s - 10) \left(s^2 - \frac{17}{3}s - \frac{10}{3} \right) = 0$$

Thus, the eigenvalues are

$$\lambda_1 = +10, \quad \lambda_{2,3} = \frac{17}{6} \pm \frac{\sqrt{409}}{6}$$

The system is unstable, and one of the eigenvalues associated with unstable modes is $\lambda_1 = +10$, as obtained by analysing the topology of the block scheme.

Note: you can determine the second unstable state mode only by computing the eigenvalues of the A matrix, i.e., only by computing a representation of the overall system in state-space equations. You cannot determine it just by relying on the analysis of the block scheme topology.

EXERCISE 2 – A SOLUTION PROPOSAL

Let's analyse the information regarding the closed-loop performance, obtained by applying the given continuous-time controller $C(s)$, and the requirement on the maximum decrease in the phase margin: in fact, these pieces of information allow you to obtain two constraints on the sampling period value to be chosen.

First useful information: recall the property of the complementary sensitivity function $F(s)$ about the bandwidth of the frequency response $F(j\omega)$ (refer to Part 9, page 40 and Part 10, page 59 of the course material)

- Usually, the transfer function $F(s)$ acts as a **low-pass filter**, with bandwidth

$$B \simeq [0, \omega_c]$$

where ω_c is the crossover angular frequency associated with the open-loop transfer function $L(s)$ [refer to Part 9, pages 10-12 for the definitions of both the transfer functions $L(s)$ and $F(s)$].

- Choice of the sampling angular frequency ω_s based on the closed-loop bandwidth: according to the empirical rule reported in Part 10, page 60 of the course material, a good choice of ω_s must obey the following inequalities set

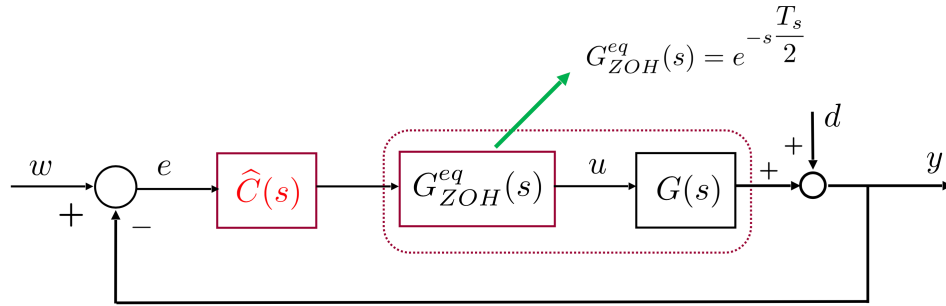
$$\omega_s : \alpha \omega_c < \omega_s < 10 \alpha \omega_c, \quad \alpha \in [5, 10]$$

- The first constraint on the sampling period: according to the just recalled empirical rule, the sampling period T_s must obey the following inequalities set:

$$\frac{2\pi}{10\alpha\omega_c} < T_s < \frac{2\pi}{\alpha\omega_c}, \quad \alpha \in [5, 10]$$

Second useful information: recall the approximated formulation reported in Part 10, pages 62-65 of the course material

- The combined effect of sampling and applying a Zero Order Hold (ZOH) is roughly equivalent to the insertion of a delay term



if

$$\omega_c \ll \frac{\omega_s}{2}, \quad \omega_s = 2\pi \frac{1}{T_s} \implies T_s \ll \frac{\pi}{\omega_c}$$

- The second constraint on the sampling period T_s : exploiting the approximating delay term, you can efficiently compute the decrease in the phase margin due to the sampling and the presence of the ZOH:

$$\delta_\varphi = -\frac{\omega_c T_s}{2} \cdot \frac{180^\circ}{\pi} \implies |\delta_\varphi| \leq \bar{\delta} \implies T_s \leq \frac{\bar{\delta}}{\omega_c} \cdot \frac{2\pi}{180^\circ}$$

The constraints on the sampling period: in conclusion, the sampling period T_s to be chosen should obey the following inequalities

$$\begin{cases} \frac{2\pi}{10\alpha\omega_c} < T_s < \frac{2\pi}{\alpha\omega_c}, & \alpha \in [5, 10] \\ T_s \leq \frac{\bar{\delta}}{\omega_c} \cdot \frac{2\pi}{180^\circ} \end{cases}$$

Important remark: usually, the last inequality is tighter than the first set of inequalities. The minimum value for the sampling period, as declared in the first set of inequalities, is only a good practice advice, not a mathematical rule. It's only advice to avoid a too small sampling period (thus a too high sampling frequency) if not necessary.

Choosing the sampling period:

$$\begin{cases} \omega_c = 4.5 \text{ rad/sec} \\ |\delta_\varphi| \leq 1^\circ = \bar{\delta} \end{cases} \Rightarrow \begin{cases} \left(\frac{2\pi}{100\omega_c} < \right) T_s < \frac{2\pi}{5\omega_c} \\ T_s \leq \frac{1^\circ}{\omega_c} \cdot \frac{2\pi}{180^\circ} \end{cases} \Rightarrow \begin{cases} (1.396 \cdot 10^{-2} <) T_s < 2.792 \cdot 10^{-1} \\ T_s \leq 7.757 \cdot 10^{-3} \end{cases}$$

In the end, to meet the requirement for the maximum decrease in the phase, the sampling period must obey the inequality:

$$T_s \leq 7.757 \cdot 10^{-3} \text{ sec.}$$

Thus, an appropriate choice can be as follows:

$$\bar{T}_s = 1.0 \cdot 10^{-3} \text{ sec.}$$

Digital implementation of the controller, using the Tustin approximate transformation: applying the chosen sampling period $\bar{T}_s$, the Tustin transformation formula becomes

$$\bar{T}_s = 1.0 \cdot 10^{-3} \text{ sec.} \Rightarrow \text{Tustin transformation: } s = \frac{2}{T_s} \left(\frac{z-1}{z+1} \right) = 2000 \left(\frac{z-1}{z+1} \right)$$

Thus, the transformed transfer function of the controller is

$$\begin{aligned} \hat{C}_{TU}(z) &= 5.0 \cdot \frac{1+s}{(81+18s+s^2)} \Big|_{s=2000\left(\frac{z-1}{z+1}\right)} = 5.0 \cdot \frac{1+2000\left(\frac{z-1}{z+1}\right)}{\left[81+18 \cdot 2000\left(\frac{z-1}{z+1}\right) + \left(2000\frac{z-1}{z+1}\right)^2\right]} \\ &= 5.0 \cdot \frac{\frac{(z+1)+2000(z-1)}{z+1}}{\left[\frac{81(z+1)^2+3.6 \cdot 10^4(z-1)(z+1)+4 \cdot 10^6(z-1)^2}{(z+1)^2}\right]} \\ &= 5.0 \cdot \frac{(z+1)(2001z-1999)}{(2009z-1991)^2} \\ &= \frac{10005}{4036081} \frac{(z+1)\left(z-\frac{1999}{2001}\right)}{\left(z-\frac{1991}{2009}\right)^2} \simeq 2.479 \cdot 10^{-3} \frac{(z+1)(z-0.999)}{(z-0.991)^2} \end{aligned}$$

Digital implementation of the controller - The algorithm:

Given the transformed transfer function of the controller $\hat{C}_{TU}(z)$, exploiting the properties of the Z-transform (recall Part 10, pages 53-56 of the course material), you can write down the difference equation implementing the discretised control law:

- First step: rewrite the discretised transfer function of the controller $\hat{C}_{TU}(z)$ using the notation with negative powers of z :

$$\hat{C}_{TU}(z) = \frac{10005}{4036081} \frac{(z+1) \left(z - \frac{1999}{2001}\right)}{\left(z - \frac{1991}{2009}\right)^2} \implies \hat{C}_{TU}(z) = \frac{10005}{4036081} \frac{(1+z^{-1}) \left(1 - \frac{1999}{2001} z^{-1}\right)}{\left(1 - \frac{1991}{2009} z^{-1}\right)^2}$$

- Second step: exploiting the properties of the Z-transform, determine the difference equation describing the discretised control algorithm, corresponding to the given discrete-time transfer function:

$$U(z) = \hat{C}_{TU}(z) \cdot E(z), \quad U(z) \triangleq \mathcal{Z}\{u[k]\}, \quad E(z) \triangleq \mathcal{Z}\{e[k]\}$$

Thus,

$$\begin{aligned} \left(1 - \frac{1991}{2009} z^{-1}\right)^2 U(z) &= \frac{10005}{4036081} (1+z^{-1}) \left(1 - \frac{1999}{2001} z^{-1}\right) E(z) \\ \left(1 - \frac{3982}{2009} z^{-1} + \frac{3964081}{4036081} z^{-2}\right) U(z) &= \frac{10005}{4036081} \left(1 + \frac{2}{2001} z^{-1} - \frac{1999}{2001} z^{-2}\right) E(z) \end{aligned}$$

Finally:

$$u[k] = \frac{3982}{2009} u[k-1] - \frac{3964081}{4036081} u[k-2] + \frac{10005}{4036081} e[k] + \frac{10}{4036081} e[k-1] - \frac{9995}{40360818} e[k-2]$$

The control algorithm: The resulting discretised control algorithm can be described using the following difference equation:

$$u[k] = 1.9821 u[k-1] - 0.9822 u[k-2] + 2.4789 \cdot 10^{-3} e[k] + 2.4776 \cdot 10^{-6} e[k-1] - 2.4764 \cdot 10^{-4} e[k-2]$$

where $e[k]$ is the sample of the error variable at the k -th sampling instant, whereas $u[k]$ is the sample of the control variable at the same time instant.